Wage Transparency and Within-Rank Salary Compression: Evidence from UC Faculty Compensation, 2012–2024

Joshua Kenworthy Agnibha Bhattacharya Jackson Wurzer

March 2026

Abstract

We examine whether the 2022 increase in the salience of wage information in the University of California (UC) system corresponds with measurable changes in within-rank faculty salary dispersion. Using a 13-year repeated cross-section of approximately 229,000 salary records from 2012 to 2024—covering five faculty ranks across ten UC campuses—we construct annual coefficients of variation (CV) for each rank and estimate log-linear OLS regressions with rank fixed effects, campus fixed effects, and a linear time trend. We find that mean nominal salaries grew by roughly 52 % over the sample period before declining 6.4 % in 2024. The baseline regression yields a *post2022* coefficient of -0.063 ($p < 0.001$), indicating wages were approximately 6.3 % below the secular trend after 2022 conditional on rank and campus. The CV declined to varying degrees for Full, Associate, Research, and Clinical professors after 2022, with Associate professors showing the largest drop of approximately 4.6 percentage points. Assistant professors are an exception: their CV rose by roughly 1.4 percentage points, consistent with external labor-market pressures. Levene's tests confirm that absolute variance increased

significantly across all ranks post-2022, but this is reconcilable with CV compression because mean salaries rose faster than the absolute spread. A difference-in-differences (DiD) interaction model finds no significant narrowing of the wage gap between Full and Assistant professors, though Associate and Clinical professors fared relatively better in the post-period. We interpret these patterns as consistent with transparency-driven internal equity adjustments concentrated within, rather than across, faculty ranks.

Contents

1 Introduction

The University of California (UC) system is one of the largest public employers in the United States, employing over 265,000 faculty and staff across ten campuses (University of California, 2025). Compensation within the UC system has long been subject to a high degree of transparency. Following a 2008 court decision, the *Sacramento Bee* created a searchable online database that allowed the public to view the salary of any California state employee, including UC faculty (Card et al., 2012, p. 2982). Since that time, multiple websites have made UC salary data easily accessible, enabling employees to compare their compensation with that of their peers on an ongoing basis.

This institutional setting provides a useful environment to study the relationship between wage transparency and wage compression. Economic theory predicts that when workers can observe peer compensation, within-group pay disparities become salient and can generate dissatisfaction, reduce effort, or increase turnover among lower-paid employees (Card et al., 2012, p. 2981). Employers may therefore face incentives to narrow wage gaps within comparable job categories in order to retain workers and maintain productivity (Cullen & Pakzad-Hurson, 2022).

Our paper focuses on the period surrounding 2022, which represents a meaningful shift in the compensation and transparency environment for UC faculty. Two developments converged: first, California's Pay Transparency Act took effect in January 2023, expanding wage disclosure requirements across the state; second, in late 2022 the UC system experienced its largest academic labor action in history, with tens of thousands of academic workers striking over wages and working conditions and obtaining significant contract settlements. Together, these events substantially elevated the salience of wage comparisons within the UC system—even though salary data had technically been public for years.

We examine whether this period corresponds with measurable changes in within-rank salary dispersion among UC faculty. Our analysis focuses on five faculty categories: Assistant, Associate, Full, Clinical, and Research professors. The main empirical tools are: (1) annual coefficients of variation (CV) computed separately for each rank; (2) a baseline log-linear OLS regression that estimates whether log wages deviated from the secular trend after 2022; (3) Levene's tests for equality of variance across the pre- and post-2022 periods; and (4) a difference-in-differences interaction model that tests whether wage gaps between ranks changed after 2022.

The remainder of the paper proceeds as follows. Section 2 describes the theoretical framework. Section 3 describes the data sources and preprocessing. Section 4 outlines the empirical methodology. Section 5 presents the results. Section 6 discusses findings and limitations. Section 7 concludes. Figures and supplementary tables appear in the Appendix.

2 Theoretical Framework

We model the 2022 transparency and labor-action shock as an event that abruptly increased the salience of peer wage comparisons within the UC faculty labor market. The theoretical mechanism draws on two strands of the economics literature.

The first strand concerns social preferences and inequality aversion. Bandiera et al. (2005) and Card et al. (2012) document that workers who learn they are paid less than comparable peers experience a significant reduction in job satisfaction and effort. Following the utility specification in Bandiera et al. (2005), an agent's utility can be written as:

$$U_i = W_i - C(e_i) - \lambda \max(0,\ W_{\text{peer}} - W_i), \tag{1}$$

where W_i is the worker's wage, $C(e_i)$ is the cost of effort, and the final term captures the disutility—an "envy cost"—from observing a higher-paid peer. The parameter $\lambda \geq 0$ indexes the salience of wage comparisons. When compensation is opaque, $\lambda \approx 0$. The 2022 transparency shock raises λ, shifting the agent's incentive-compatibility constraints and creating pressure for the principal (the UC system) to reduce within-group wage dispersion in order to prevent a decline in aggregate effort.

The second strand concerns the equilibrium effects of wage transparency. Cullen & Pakzad-Hurson (2022) show that transparency fundamentally alters bargaining dynamics by making individualized pay differences highly visible, and consequently costly for firms to sustain. In equilibrium, employers facing transparent wage structures rationally choose to reduce within-group dispersion to limit internal dissatisfaction. An important boundary condition in the Cullen–Pakzad-Hurson framework is that this compression is primarily an internal equity response: it operates within peer groups defined by comparable job classification, not across structurally different positions.

Applied to the UC setting, this framework generates two predictions. For established faculty ranks (Full, Associate, Clinical, Research), where salaries are largely set through internal processes and promotion ladders, we expect the transparency shock to induce compression: the CV should fall and, potentially, rank-level wages should lie below their pre-trend trajectory. For entry-level Assistant professors, however, salaries are primarily determined by competitive external hiring, so the UC system has limited ability to compress wages without risking recruitment failure. We therefore predict little or no compression at the bottom of the hierarchy.

3 Data

3.1 Sources

Our salary data come from two public administrative databases. Data for 2012 are drawn from the UC Annual Wage Database maintained by the UC Office of the President (`https://ucannualwage.ucop.edu/wage/`). Data for 2013 through 2024 are drawn from the California State Controller's Government Compensation in California database (`https://publicpay.ca.gov/`). All data are publicly accessible and contain no individual identifiers.

The merged panel covers 13 annual snapshots (2012–2024) and, after preprocessing, contains 229,386 professor-year observations with positive total wages, and 225,412 observations with positive regular pay.

3.2 Preprocessing and Sample Construction

Each annual file was loaded and standardized to five core columns: year, employer name, position title, regular pay, and total wages. We retained only records whose position title matched a "professor" pattern using a case-insensitive regular expression that captures titles containing "Prof" followed by a space or hyphen, thereby including all professor variants while excluding "Professional" administrative staff.

We then mapped free-text position titles into five faculty ranks using a priority-ordered keyword rule: a position is classified as *Assistant* if the title contains "assistant" or "asst"; *Associate* if it contains "associate" or "assoc"; *Clinical* if it contains "clinical" or "clin"; *Research* if it contains "research" or "res"; and *Full* otherwise (including bare "Professor" titles). This within-rank classification is central to our identification strategy: comparing an entry-level Assistant Professor to a tenured Full Professor would confound structural position differences with salary inequality. By analyzing dispersion separately within each rank, we isolate within-peer-group variance.

Before finalizing the panel, we removed duplicate rows, dropped records with negative or zero pay values, and verified that all five required columns were non-missing. The resulting analysis dataset is a *repeated cross-section*: annual CSVs contain no individual identifiers, so we cannot track the same professor across years. Each observation is an independent salary record for whoever held a given position in a given year.

3.3 Descriptive Statistics

Table 1 summarizes the annual sample size and mean salary. The total number of professor observations grew from 16,017 in 2012 to a peak of 20,176 in 2023 before declining to 18,898 in 2024. Figure 1 plots the level and distributional evolution of nominal total wages.

Table 1: Year-by-Year Professor Observations and Mean Total Wages, 2012–2024

Year	N	Mean ($)	Median ($)	Std. Dev. ($)	YoY %
2012	16,017	162,636	138,400	117,478	—
2013	15,711	172,838	146,505	122,877	+6.3%
2014	16,042	179,202	152,000	128,030	+3.7%
2015	16,508	187,178	158,688	133,957	+4.5%
2016	17,176	195,355	165,121	142,233	+4.4%
2017	17,739	202,788	171,252	148,112	+3.8%
2018	18,151	210,399	178,817	151,300	+3.8%
2019	17,963	218,634	185,050	157,137	+3.9%
2020	19,224	225,799	195,000	154,026	+3.3%
2021	19,415	237,191	201,700	166,500	+5.0%
2022	19,737	249,471	211,045	174,503	+5.2%
2023	20,176	264,251	224,633	186,085	+5.9%
2024	18,898	247,276	208,691	179,869	−6.4%

Source: UC Annual Wage Database (2012) and California State Controller's Office (2013–2024). Sample restricted to professor positions with positive total wages.

Mean nominal wages grew steadily from $162,636 in 2012 to a peak of $264,251 in 2023—a cumulative increase of approximately 52 %, or roughly 4 % per year on average. Year-over-year growth was remarkably stable from 2014 through 2021, ranging from 3.3 % to 5.0 % annually, suggesting a consistent underlying wage-setting regime during the pre-period. The acceleration in 2022 (+5.2 %) and 2023 (+5.9 %) stands out modestly above that trend, as does the sharp reversal in 2024 (−6.4 %).

Across ranks in 2024, Clinical faculty hold the highest mean compensation at approximately $371,264 for the rank overall, reflecting the medical compensation structure at UC health science campuses. Full Professors earn a mean of \$261,329, Associate Professors \$233,613, and Assistant Professors \$183,209. Among campuses, UC San Francisco (\$298,740) and UC San Diego (\$292,983) report the highest mean total wages, while UC Riverside (\$167,993) and UC Merced (\$164,933) report the lowest.

All analysis uses nominal wages. No inflation adjustment is applied; the analysis focuses on within-period distributional patterns rather than real purchasing power comparisons.

4 Empirical Strategy

4.1 Measuring Within-Rank Wage Dispersion

Our primary measure of wage compression is the coefficient of variation (CV), defined as the ratio of the standard deviation to the mean within a rank-year cell:

$$\mathrm{CV}_{r,t} = \frac{\sigma_{r,t}}{\mu_{r,t}} \times 100, \tag{2}$$

where r indexes rank and t indexes year. The CV normalizes dispersion by the mean level, allowing meaningful comparison across periods with different average salary levels. Because nominal wages rose substantially over the sample period, a raw comparison of standard deviations would be uninformative; the CV controls for this scale effect. We also report the standard deviation and interquartile range (IQR) as supplementary measures.

4.2 Baseline OLS Regression

To estimate the conditional association between the post-2022 period and log wages, we estimate:

$$\log(\mathrm{TotalWages}_i) = \beta_0 + \beta_1 \cdot \mathrm{post2022}_i + \beta_2 \cdot \mathrm{Rank}_i + \beta_3 \cdot \mathrm{Campus}_i + \beta_4 \cdot \mathrm{Year}_i + \varepsilon_i, \tag{3}$$

where post2022 is an indicator equal to 1 for observations from 2022 onward and 0 otherwise. Rank and Campus enter as sets of dummy variables (fixed effects), with Assistant Professor and UC Berkeley as the respective reference categories. Year is a continuous linear time trend that controls for secular growth in nominal wages. The coefficient β_1 captures whether log wages deviated from the underlying trend after 2022, conditional on rank and campus. Using a log transformation on the outcome allows coefficients to be interpreted as approximate percentage changes.

Because the wage distribution is right-skewed and exhibits heteroskedasticity—variance among Full Professors is substantially wider than among Assistant Professors—we use heteroskedasticity-robust (HC1) standard errors throughout. We verify that multicollinearity between the continuous Year trend and the binary post2022 step function is not severe by computing Variance Inflation Factors (VIF).

As a robustness check, we re-estimate Equation (3) with log(RegularPay) as the dependent variable. Regular pay reflects only base compensation, excluding bonuses and other discretionary pay. Consistency across both outcomes confirms that findings are not an artifact of variation in bonus structures.

We also run the same specification separately within each rank—the *within-rank regressions*—which hold rank composition constant by construction and test whether the post-2022 level shift persists inside individual faculty categories.

4.3 Levene's Test for Variance Equality

We test whether the variance of salaries changed significantly between the pre-2022 period (2012–2021) and the post-2022 period (2022–2024) within each rank using Levene's test for equality of variances. Levene's test is robust to non-normal distributions, making it appropriate for wage data. The variance ratio (post-2022 variance divided by pre-2022 variance) indicates the direction of change: a ratio greater than 1 indicates that absolute variance *increased* after 2022, while a ratio less than 1 indicates it decreased.

4.4 Difference-in-Differences Interaction Model

To test whether wage gaps *between* ranks changed after 2022—the strongest form of the compression hypothesis—we estimate:

$$\log(\text{TotalWages}_i) = \gamma_0 + \gamma_1 \cdot \text{post2022}_i \times \text{Rank}_i + \gamma_2 \cdot \text{Campus}_i + \gamma_3 \cdot \text{Year}_i + \varepsilon_i. \quad (4)$$

The interaction terms post2022 × Rank capture whether the post-2022 wage shift was differentially larger or smaller for a given rank relative to the reference category (Assistant Professor). A negative interaction for Full Professors would indicate that Full Professors experienced a larger proportional decline than Assistant Professors after 2022, consistent with compression from the top of the distribution.

We note an important limitation of this design: our data are a repeated cross-section with no individual identifiers. The DiD model therefore captures changes in the *distribution* of wages across groups, not individual wage trajectories. We cannot rule out unobserved compositional shifts within ranks—for example, if the mix of specializations or seniority levels within a rank changed after 2022.

5 Results

5.1 Salary Trends

Figure 1 displays the evolution of mean total wages, year-over-year growth, quartile salary trends, and total professor headcount from 2012 to 2024. Mean nominal wages grew steadily and monotonically from 2012 through 2023, with the 2024 observation representing the first nominal decline in the sample period (−6.4 %). The IQR widened over time, reflecting growing absolute dispersion, but the rate of widening decelerated after 2022 for most ranks. The professor headcount grew from roughly 16,000 in 2012 to just over 20,000 in 2023, with a decline to approximately 18,900 in 2024.

Figure 4 shows headcount by rank over time. The Full Professor category is the largest throughout the sample. The Clinical Professor category grew substantially, from roughly 1,400 observations in 2012 to over 2,500 by 2024, reflecting expanded health-system activity across UC campuses.

5.2 Within-Rank Coefficient of Variation

Figure 5 and Figure 2 report the CV for each rank and year. The average pre-2022 CV (2012–2021 mean) and post-2022 CV (2022–2024 mean) for each rank are as follows:

Table 2: CV Summary: Pre-2022 vs. Post-2022 Average

Rank	Avg CV Pre-2022 (%)	Avg CV Post-2022 (%)	Change (pp)
Assistant	73.13	74.55	+1.42
Associate	76.52	71.90	−4.61
Full	59.22	56.30	−2.92
Clinical	64.81	64.19	−0.63
Research	68.53	65.64	−2.89

Pre-2022 = average CV across 2012–2021; Post-2022 = average CV across 2022–2024. CV computed as $100 \times \sigma/\mu$ within each rank-year cell. Source: notebook cell 36 (deduplicated regression panel, N = 229,386).

Four of the five ranks show a decline in relative wage dispersion after 2022. The largest drop occurs among Associate Professors (−4.61 percentage points), followed by Full Professors (−2.92 pp) and Research Professors (−2.89 pp). Clinical Professors exhibit a modest decline of −0.63 pp. Assistant Professors are the notable exception: their CV increased by +1.42 pp, indicating that relative wage dispersion among entry-level faculty *widened* after 2022.

The CV trends for Full and Associate Professors show a broadly monotonic decline from 2012 onward that predates the 2022 threshold, suggesting that within-rank convergence was a gradual ongoing process rather than a discrete policy response. The 2022 threshold does not appear to coincide with a sudden break for most ranks. The Assistant Professor CV, in contrast, declined through 2020 before reversing upward in 2022–2024.

5.3 Levene's Test for Variance Equality

Figure 9 reports Levene's test results. Across all five ranks, the null hypothesis of equal variances is rejected at $p < 0.001$, indicating that the variance of salaries changed significantly between the pre- and post-2022 periods. However, the variance *ratios* range from 1.32 (Full) to 1.59 (Assistant), all greater than 1, indicating that absolute salary variance *increased* post-2022 across every rank.

This result appears to contradict the CV compression finding but is fully reconcilable. Mean salaries rose substantially during the post-2022 period, driven by inflation, secular wage growth, and the consequences of the 2022 labor action. Because mean salaries grew faster than the absolute spread of the distribution for most ranks, relative dispersion (as measured by the CV) fell even as the dollar-denominated spread widened. The appropriate measure for assessing the compression hypothesis is therefore the CV, not the raw variance: workers experience inequality in proportional terms relative to their peers' wages, not in absolute dollar differences.

5.4 Baseline OLS Results

Figure 6 presents the baseline OLS results for log(TotalWages). The post-2022 coefficient is -0.0634 (SE = 0.0063, $p < 0.001$), indicating that log total wages were approximately 6.3 % below the level predicted by the secular trend after 2022, conditional on rank and campus. All rank fixed effects are positive and strongly significant relative to the Assistant Professor baseline, confirming a clear compensation hierarchy: Clinical > Full > Research > Associate > Assistant. The linear time trend coefficient of 0.0475 implies annual nominal wage growth of roughly 4.9 % per year along the pre-trend.

Figure 7 presents the robustness check using log(RegularPay) as the dependent variable. The post-2022 coefficient is -0.0410 (SE $= 0.0043$, $p < 0.001$). The smaller magnitude relative to the total-wages result suggests that some of the post-2022 decline in total wages reflects reduced variable pay (bonuses, overload pay), but the direction and significance of the finding are unchanged. Both models use the deduplicated panel of 229,386 and 225,412 observations respectively.

5.5 Multicollinearity Check

Figure 10 reports the Variance Inflation Factors (VIF) for the baseline regression predictors. The VIF for Year is 2.197 and for post2022 is 2.193, both well below the commonly used concern threshold of 5 and far below the severe threshold of 10. The VIF for rank and campus are approximately 1.0. Multicollinearity is not a concern in the baseline specification.

5.6 Difference-in-Differences Interaction Results

Figure 8 presents the interaction regression results. The baseline post2022 term is -0.0813 (SE $= 0.0103$, $p < 0.001$), reflecting the post-period wage decline for Assistant Professors (the reference group).

The interaction term for Full Professors is -0.0016 ($p = 0.888$), which is economically negligible and statistically indistinguishable from zero. This means Full Professors experienced essentially the same post-2022 wage shift as Assistant Professors, providing no evidence of between-rank compression from the top.

Associate Professors show a statistically significant positive interaction (coef $= +0.0476$, $p = 0.0002$), indicating they fared *better* relative to Assistant Professors after 2022. This is broadly consistent with the strong CV compression observed among Associates: their wages held up better than those of either Assistant or Full Professors in the post-period. Clinical Professors also show a positive interaction (coef $= +0.0548$, $p = 0.002$), reflecting continued compensation growth in health-system faculty markets. The Research Professor interaction is positive but statistically insignificant (coef = +0.0301, $p = 0.253$).

Taken together, the DiD evidence does not support the hypothesis that the UC systematically narrowed the wage gap between senior and junior faculty after 2022 through targeted compression at the top.

5.7 Within-Rank Regressions

Figures 11–15 report the within-rank baseline regressions, which hold rank composition constant and directly test whether post-2022 wages deviated from the within-rank secular trend. Results are summarized in Table 3.

Table 3: Within-Rank Post-2022 Coefficient Summary

Rank	Coef.	Std. Err.	t	p-value	Approx. Effect
Assistant	−0.0954	0.0132	−7.23	< 0.001	−9.1%
Associate	−0.0452	0.0130	−3.47	0.001	−4.4%
Full	−0.0728	0.0086	−8.49	< 0.001	−7.0%
Clinical	−0.0272	0.0235	−1.16	0.247	−2.7% (n.s.)
Research	+0.0092	0.0375	+0.25	0.805	+0.9% (n.s.)

Specification: $\log(\text{TotalWages}) \sim \text{post2022} + C(\text{EmployerName}) + \text{Year}$, HC1 robust SE. Approximate percentage effect computed as $(\exp(\text{coef}) - 1) \times 100$.

The within-rank results reveal an important pattern: the post-2022 wage shortfall relative to trend is largest for Assistant Professors (−9.1%, $p < 0.001$) and Full Professors (−7.0%, $p < 0.001$), while it is smaller and insignificant for Clinical and Research professors. Associate Professors show a moderate, significant decline of −4.4%. The fact that Assistant Professors show the largest within-rank decline in absolute magnitude, despite their CV rising, reflects that the *level* of their wages fell relative to trend while the *spread* of their distribution widened—consistent with the external market setting floor wages that vary widely across disciplines.

5.8 Pre-versus Post-2022 Salary Distributions

Figure 3 displays overlapping salary histograms for each rank, comparing the pre-2022 pooled distribution to the post-2022 pooled distribution. For Full and Associate Professors, the post-2022 distribution shifts rightward, reflecting higher mean wages, while the relative spread of the distribution narrows modestly. For Assistant Professors, the rightward shift is present but the tails of the distribution widen proportionally, consistent with the CV increase for that rank. Clinical Professors show a broad rightward shift with a pronounced right tail.

6 Discussion

6.1 Interpreting the Main Findings

The results present a nuanced picture of how the UC faculty labor market responded to the 2022 transparency and labor-action shock. Two findings stand out.

First, there is clear evidence of *within-rank* relative wage compression among established faculty. The CVs for Full, Associate, Research, and Clinical professors all declined after 2022, with Associate professors exhibiting the strongest signal ($-4.61\,\text{pp}$). This pattern is consistent with the theoretical prediction in Cullen & Pakzad-Hurson (2022): when wage comparisons become more salient, employers face pressure to narrow within-group pay gaps to limit dissatisfaction among employees who directly compare their compensation with same-rank peers.

Second, there is no evidence of *between-rank* compression. The DiD interaction for Full versus Assistant Professors is economically small and statistically indistinguishable from zero. This is theoretically coherent: inequality aversion operates among direct peers, not across structurally distinct positions. An Assistant Professor has no basis for comparing their salary to a Full Professor's, given the evident differences in seniority, tenure, and accumulated human capital. The utility specification in Equation (1) implies that λ is activated by comparisons within a relevant reference group, not across all employees.

The exception among ranks—Assistant Professors—highlights an important boundary condition on transparency-driven compression. Entry salaries are set through a competitive national academic job market with substantial cross-disciplinary variation, and the UC cannot compress entry wages without risking recruitment failures in high-demand fields. The $+1.42$ pp increase in the Assistant Professor CV is consistent with growing salary heterogeneity in external markets for new faculty.

6.2 Reconciling CV and Levene Results

Levene's tests show that absolute salary variance increased significantly after 2022 across all ranks, which superficially contradicts the CV-based compression finding. The reconciliation is straightforward. Mean salaries rose substantially during the post-2022 period, driven by inflation, secular wage growth, and the historic 2022 academic strikes. Because the mean grew faster than the standard deviation for senior ranks, the CV declined even as the dollar-denominated spread of the distribution widened. This illustrates why the CV is the appropriate dispersion metric for the compression hypothesis: workers experience inequality in proportional terms relative to their peers' compensation levels, not as an absolute dollar gap.

6.3 Clinical Professors

The positive and significant DiD interaction for Clinical Professors (coef = $+0.0548$, $p = 0.002$) indicates they fared better than Assistant Professors in the post-2022 period. This likely reflects continued growth in health-system compensation at UC's medical campuses, driven by broader dynamics in the physician and academic medicine labor market. Clinical faculty operate under different compensation structures than academic-track faculty, limiting the degree to which internal UC equity norms constrain their pay.

6.4 The 2024 Decline

The -6.4% year-over-year decline in mean total wages in 2024 is the most pronounced single-year change in the sample. At the same time, total observations fell from 20,176 in 2023 to 18,898 in 2024, a decline of approximately 6.3%. Because our data are a repeated cross-section, we cannot determine the extent to which the mean wage decline reflects actual pay cuts versus compositional turnover—specifically, whether higher-paid senior faculty exited the UC system while lower-paid new hires entered. The 2024 pattern warrants cautious interpretation.

6.5 Limitations

Repeated cross-section. The fundamental limitation of this study is the absence of individual identifiers. We cannot track any professor across years, so we cannot measure individual wage growth, distinguish pay cuts from turnover, or control for unobserved compositional shifts within ranks. It is possible that apparent within-rank compression among Full Professors reflects adverse selection: if the UC compressed wages by capping high earners, and those individuals subsequently departed for private universities, the observed compression in the remaining sample would reflect attrition rather than deliberate wage-setting policy.

Identification. Our design compares the post-2022 period to the pre-2022 secular trend. We cannot claim that any observed change *caused* by the transparency mandate or the strikes, as multiple forces changed simultaneously in 2022–2023. The post-2022 indicator absorbs any event occurring at that threshold. We therefore frame our findings as descriptive associations rather than causal estimates.

Nominal wages only. All analysis uses nominal wages. Without verified inflation data, we do not convert to real wages. Over a 13-year period spanning significant inflation, the nominal trend likely overstates real wage growth. However, our main finding concerns *relative* within-rank dispersion (the CV), which is unaffected by common inflation because it divides the standard deviation by the mean.

7 Conclusion

This paper examined within-rank salary dispersion among University of California faculty using 13 years of administrative compensation data spanning 2012–2024. Our analysis focused on whether the period surrounding the 2022 California Pay Transparency Act and the historic UC academic labor strikes corresponds with measurable changes in the distribution of wages within faculty ranks.

Our main findings are as follows. Mean nominal faculty wages grew by approximately 52 % over the sample period before declining 6.4 % in 2024. The coefficient of variation declined after 2022 for Full (−2.9 pp), Associate (−4.6 pp), Research (−2.9 pp), and Clinical (−0.6 pp) professors, indicating a narrowing of relative pay disparities within these ranks. Assistant Professor CV increased by +1.4 pp, consistent with external labor-market pressures that prevent internal compression at the hiring margin. The baseline OLS regression finds that log wages were approximately 6.3 % below the secular trend after 2022, conditional on rank and campus. The DiD interaction model finds no evidence that wage gaps between Full and Assistant Professors narrowed, but Associate and Clinical professors fared relatively better in the post-period. Levene's tests confirm that absolute variance increased across all ranks, but this is consistent with CV compression because mean salaries grew faster than the spread for senior ranks.

These findings suggest that wage transparency and labor actions can induce relative salary compression *within* established peer groups but have a more limited impact *across* structurally distinct positions. Internal equity pressures appear to operate primarily among direct peers who observe and compare their wages, while the compensation of entry-level faculty remains tethered to external market conditions. Future research with individual-level panel data would be needed to separate deliberate wage-setting compression from compositional turnover effects.

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Appendix A: Mathematical Model of Inequality Aversion

To formalize the theoretical mechanism, we employ the inequality aversion framework from Bandiera et al. (2005). The agent's utility function is:

$$U_i = W_i - C(e_i) - \lambda \max(0,\ W_{\text{peer}} - W_i), \tag{5}$$

where W_i is the worker's wage, $C(e_i)$ is the convex cost of effort e_i, and the final term captures the disutility from observing a higher-paid peer, scaled by the salience parameter $\lambda \geq 0$.

Prior to 2022, UC salary data were publicly posted but received limited attention; we model this as $\lambda \approx 0$, so pay comparisons generated negligible disutility. The 2022 events—the Pay Transparency Act, the strikes, and intensified media coverage of faculty pay—shifted $\lambda > 0$ for a meaningful share of the workforce. Once $\lambda > 0$, the incentive-compatibility constraint for effort is tightened: the principal (the UC system) must either raise W_i toward W_{peer} or reduce W_{peer} toward W_i to maintain effort levels. Both channels narrow the distribution. The result is that the optimal contract involves lower within-group wage variance, i.e., wage compression.

The prediction is limited to comparisons within a relevant reference group. An Assistant Professor does not compare their wage to a Full Professor's because the rank difference is institutionally recognized and perceived as structurally justified. The model therefore predicts compression within ranks but not across ranks—consistent with our empirical findings.

Appendix B: Figures

Figure 1: UC Professor Salary Trends, 2012–2024 (Nominal). Top left: mean total wages with ±1 standard deviation band. Top right: year-over-year percentage change in mean salary; the 2024 bar (−6.4 %) is the only negative value in the sample. Bottom left: quartile salary trends (Q25, median, Q75). Bottom right: total professor observation count by year. Vertical dashed line marks the 2022 threshold.

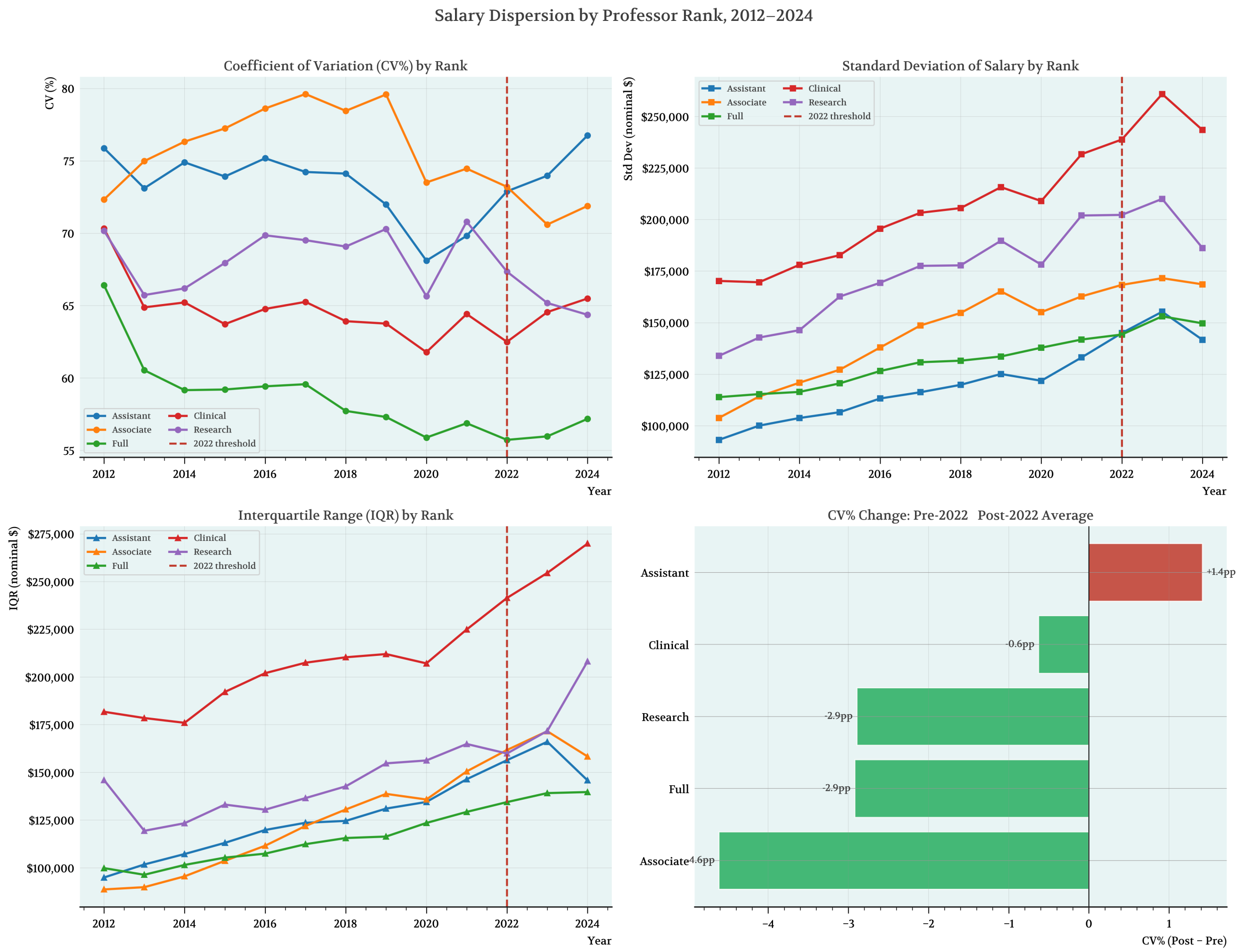

Figure 2: Salary Dispersion by Professor Rank, 2012–2024. Top left: coefficient of variation (CV%) by rank; lower values indicate less relative dispersion. Top right: standard deviation of salary by rank. Bottom left: interquartile range by rank. Bottom right: bar chart summarizing the CV change in percentage points (post-2022 minus pre-2022 average). Red bar indicates CV expansion (Assistant); green bars indicate CV compression. Vertical dashed line marks the 2022 threshold.

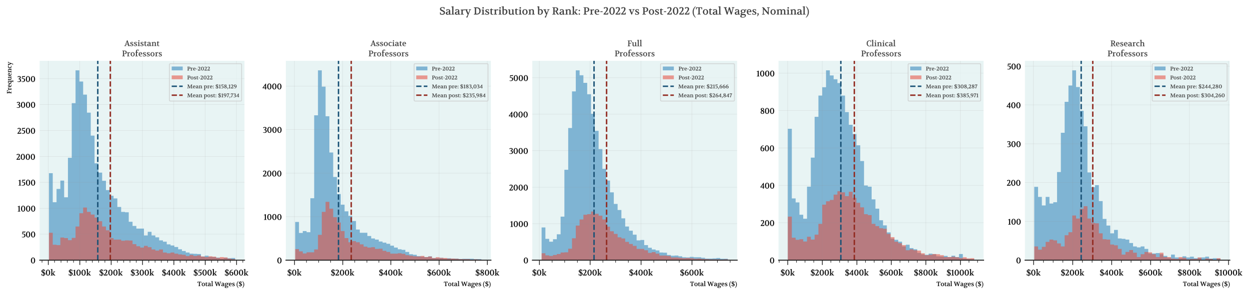

Figure 3: Salary Distribution by Rank: Pre-2022 vs. Post-2022 (Total Wages, Nominal). Each panel overlays the pooled pre-2022 (blue) and post-2022 (red) salary histograms for one rank. Vertical dashed lines mark the mean for each period. For Full and Associate Professors, the post-2022 distribution shifts rightward while the relative spread narrows modestly. For Assistant Professors, the distribution widens proportionally with the rightward shift.

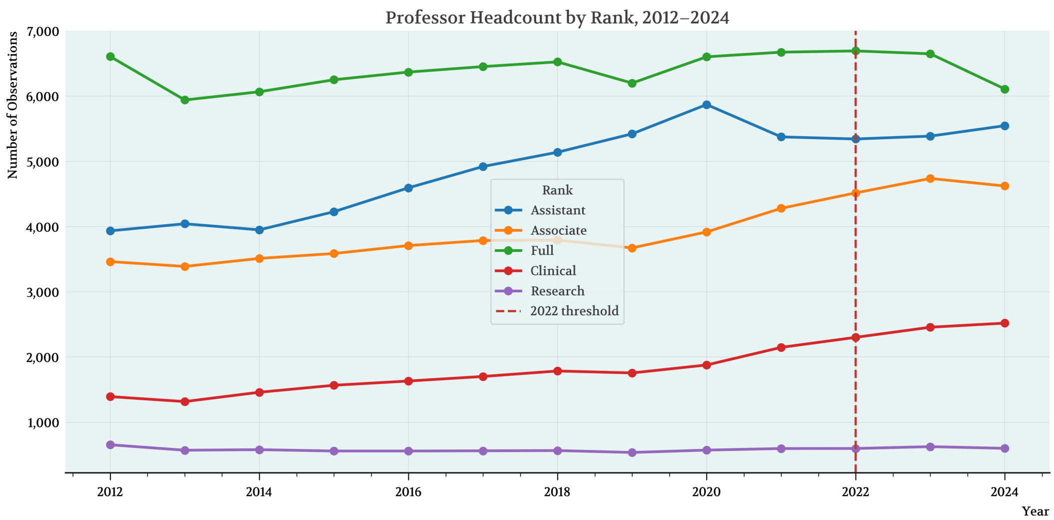

Figure 4: Professor Headcount by Rank, 2012–2024. Full Professors (green) are consistently the largest group. Clinical Professors (red) grew substantially from roughly 1,400 in 2012 to over 2,500 by 2024. Assistant Professors (blue) grew through 2019–2020 before declining in 2023–2024. Vertical dashed line marks the 2022 threshold.

Appendix C: Tables

Table: Coefficient of Variation (CV%) by Rank and Year

Year	Assistant	Associate	Full	Clinical	Research
2012	75.88	72.34	66.42	70.33	70.17
2013	73.12	74.99	60.54	64.89	65.74
2014	74.90	76.33	59.17	65.22	66.20
2015	73.92	77.25	59.21	63.73	67.95
2016	75.19	78.62	59.43	64.78	69.86
2017	74.24	79.62	59.58	65.27	69.53
2018	74.13	78.46	57.73	63.93	69.09
2019	71.99	79.59	57.31	63.76	70.30
2020	68.11	73.51	55.89	61.79	65.66
2021	69.83	74.47	56.88	64.42	70.81
2022	72.90	73.21	55.73	62.50	67.35
2023	73.98	70.60	55.97	64.56	65.19
2024	76.76	71.89	57.19	65.50	64.37

Coefficient of Variation (CV%) = 100 × (Std Dev / Mean). Computed from deduplicated panel.
Sample: professors with TotalWages > 0, years 2012–2024.

Figure 5: Coefficient of Variation (CV%) by Rank and Year, 2012–2024. Computed as $100 \times \sigma/\mu$ within each rank-year cell on the deduplicated panel. Sample restricted to professors with TotalWages > 0.

Table 1. Baseline OLS: log(Total Wages) ~ post2022 + Rank FE + Campus FE + Year

Variable	Coef.	Std. Err.	t-stat	p-value	95% CI
Associate***	0.2318	0.0057	40.983	0.0000	[0.2207, 0.2429]
Clinical***	0.6329	0.0084	75.106	0.0000	[0.6164, 0.6494]
Full***	0.5063	0.0049	103.550	0.0000	[0.4967, 0.5159]
Research***	0.4455	0.0119	37.518	0.0000	[0.4222, 0.4687]
post2022***	-0.0634	0.0063	-10.010	0.0000	[-0.0759, -0.0510]
Campus FE	Yes				
Year Trend	Yes				

N = 229,386 R^2 = 0.1024
Reference category: Assistant Professor.
HC1 heteroskedasticity-robust standard errors.   Stars: *** $p<0.001$ ** $p<0.01$ * $p<0.05$

Figure 6: Table 1: Baseline OLS regression of log(Total Wages) on post2022, rank fixed effects, campus fixed effects, and a linear year trend. $N = 229{,}386$. HC1 heteroskedasticity-robust standard errors. Reference category: Assistant Professor.

Table 2. Baseline OLS: log(Regular Pay) ~ post2022 + Rank FE + Campus FE + Year

Variable	Coef.	Std. Err.	t-stat	p-value	95% CI
Associate***	0.2843	0.0038	75.125	0.0000	[0.2769, 0.2917]
Clinical***	0.6364	0.0052	122.608	0.0000	[0.6262, 0.6465]
Full***	0.6871	0.0035	197.242	0.0000	[0.6802, 0.6939]
Research***	0.5424	0.0091	59.335	0.0000	[0.5244, 0.5603]
post2022***	-0.0410	0.0043	-9.643	0.0000	[-0.0494, -0.0327]
Campus FE	Yes				
Year Trend	Yes				

N = 225,412 R^2 = 0.2164
Reference category: Assistant Professor.
HC1 heteroskedasticity-robust standard errors.   Stars: *** p<0.001 ** p<0.01 * p<0.05

Figure 7: Table 2: Robustness check—baseline OLS regression of log(Regular Pay) on post2022, rank fixed effects, campus fixed effects, and a linear year trend. $N = 225{,}412$. HC1 robust standard errors. Reference category: Assistant Professor.

Table 3. DiD Compression: log(Total Wages) ~ post2022 × C(Rank) + Campus FE + Year

Variable	Coef.	Std. Err.	t-stat	p-value	95% CI
Intercept***	-84.3099	1.5467	-54.510	0.0000	[-87.3414, -81.2785]
post2022***	-0.0813	0.0103	-7.857	0.0000	[-0.1016, -0.0610]
Year***	0.0475	0.0008	61.954	0.0000	[0.0460, 0.0490]
Rank FE	Yes				
Campus FE	Yes				
post2022:Rank: Associate***	0.0476	0.0129	3.681	0.0002	[0.0223, 0.0730]
post2022:Rank: Clinical**	0.0548	0.0179	3.058	0.0022	[0.0197, 0.0899]
post2022:Rank: Full	-0.0016	0.0111	-0.141	0.8876	[-0.0233, 0.0202]
post2022:Rank: Research	0.0301	0.0263	1.143	0.2531	[-0.0215, 0.0817]

N = 229,386 R^2 = 0.1026
DiD spec: log_total_wages ~ post2022 * C(Rank) + C(EmployerName) + Year.
HC1 heteroskedasticity-robust standard errors.   Stars: *** p<0.001 ** p<0.01 * p<0.05
Reference category: Assistant professors.

Figure 8: Table 3: Difference-in-differences compression test. Dependent variable: log(Total Wages). Specification includes post2022, the full interaction post2022 × Rank, campus fixed effects, and a linear year trend. $N = 229{,}386$. HC1 robust standard errors. Reference category: Assistant Professor.

Table: Levene's Test for Variance Equality — Pre-2022 vs Post-2022

Rank	Pre-2022 Variance	Post-2022 Variance	Variance Ratio	Levene Stat	p-value	Significant
Assistant	1.372e+10	2.185e+10	1.5928	591.7232	4.1563e-130	Yes
Associate	2.069e+10	2.878e+10	1.3913	353.2237	1.5528e-78	Yes
Full	1.682e+10	2.226e+10	1.3238	288.7538	1.1953e-64	Yes
Clinical	4.117e+10	6.172e+10	1.4990	194.4820	4.9729e-44	Yes
Research	2.926e+10	4.012e+10	1.3713	36.3683	1.7100e-09	Yes

Levene's test for equality of variances (pre-2022 vs post-2022) within each professor rank.
Variance Ratio > 1: variance increased post-2022. Significant at α = 0.05.

Figure 9: Levene's Test for Variance Equality—Pre-2022 vs. Post-2022. Variance Ratio > 1 indicates that absolute variance *increased* post-2022. All five ranks show statistically significant variance changes ($p < 0.001$). This result is consistent with CV compression because mean salaries rose faster than the absolute spread for senior ranks.

Table: Variance Inflation Factors (VIF) — Baseline Regression Predictors

Predictor	VIF
Year	2.197
post2022	2.193
Rank	1.005
EmployerName	1.004

VIF computed using label-encoded Rank and EmployerName (approximation).
Threshold: VIF > 5 = moderate concern; VIF > 10 = severe multicollinearity.
Model: log_total_wages ~ post2022 + C(Rank) + C(EmployerName) + Year | HC1 robust SE.

Figure 10: Variance Inflation Factors (VIF) for the baseline regression predictors. All VIFs are well below the concern threshold of 5, confirming that multicollinearity between the continuous Year trend and the binary post2022 indicator is not severe.

Within-Rank Regression: Assistant Professors
log(Total Wages) ~ post2022 + Campus FE + Year

Variable	Coef.	Std. Err.	t-stat	p-value	95% CI
Intercept***	-88.6824	3.2485	-27.299	0.0000	[-95.0493, -82.3154]
Campus FE	Yes				
post2022***	-0.0954	0.0132	-7.226	0.0000	[-0.1213, -0.0696]
Year***	0.0496	0.0016	30.779	0.0000	[0.0464, 0.0527]

N = 63,756 R^2 = 0.0531
Formula: log_total_wages ~ post2022 + C(EmployerName) + Year | Assistant professors only.
HC1 heteroskedasticity-robust standard errors.   Stars: *** p<0.001 ** p<0.01 * p<0.05

Figure 11: Within-Rank Regression: Assistant Professors. Specification: $\log(\text{TotalWages}) \sim \text{post2022} + C(\text{EmployerName}) + \text{Year}$. $N = 63{,}756$. HC1 robust standard errors.

Within-Rank Regression: Associate Professors
log(Total Wages) ~ post2022 + Campus FE + Year

Variable	Coef.	Std. Err.	t-stat	p-value	95% CI
Intercept***	-86.7557	3.1266	-27.747	0.0000	[-92.8838, -80.6276]
Campus FE	Yes				
post2022***	-0.0452	0.0130	-3.468	0.0005	[-0.0708, -0.0197]
Year***	0.0488	0.0016	31.476	0.0000	[0.0458, 0.0518]

N = 50,988 R^2 = 0.0645
Formula: log_total_wages ~ post2022 + C(EmployerName) + Year | Associate professors only.
HC1 heteroskedasticity-robust standard errors.   Stars: *** p<0.001 ** p<0.01 * p<0.05

Figure 12: Within-Rank Regression: Associate Professors. $N = 50{,}988$. Specification and standard errors as in Figure 11.

Within-Rank Regression: Full Professors
log(Total Wages) ~ post2022 + Campus FE + Year

Variable	Coef.	Std. Err.	t-stat	p-value	95% CI
Intercept***	-80.6041	2.1388	-37.687	0.0000	[-84.7960, -76.4121]
Campus FE	Yes				
post2022***	-0.0728	0.0086	-8.493	0.0000	[-0.0896, -0.0560]
Year***	0.0460	0.0011	43.361	0.0000	[0.0439, 0.0481]

N = 83,141 R^2 = 0.0523
Formula: log_total_wages ~ post2022 + C(EmployerName) + Year | Full professors only.
HC1 heteroskedasticity-robust standard errors.   Stars: *** p<0.001 ** p<0.01 * p<0.05

Figure 13: Within-Rank Regression: Full Professors. $N = 83{,}141$. Specification and standard errors as in Figure 11.

Within-Rank Regression: Clinical Professors
log(Total Wages) ~ post2022 + Campus FE + Year

Variable	Coef.	Std. Err.	t-stat	p-value	95% CI
Intercept***	-85.1453	6.1922	-13.750	0.0000	[-97.2819, -73.0088]
Campus FE	Yes				
post2022	-0.0272	0.0235	-1.157	0.2473	[-0.0732, 0.0189]
Year***	0.0477	0.0031	15.530	0.0000	[0.0417, 0.0537]

N = 23,918 R^2 = 0.0785
Formula: log_total_wages ~ post2022 + C(EmployerName) + Year | Clinical professors only.
HC1 heteroskedasticity-robust standard errors.   Stars: *** p<0.001 ** p<0.01 * p<0.05

Figure 14: Within-Rank Regression: Clinical Professors. $N = 23{,}918$. The post2022 coefficient is negative but not statistically significant ($p = 0.247$), reflecting the distinct dynamics of health-system compensation.

Within-Rank Regression: Research Professors
log(Total Wages) ~ post2022 + Campus FE + Year

Variable	Coef.	Std. Err.	t-stat	p-value	95% CI
Intercept***	-53.6338	8.7700	-6.116	0.0000	[-70.8226, -36.4449]
Campus FE	Yes				
post2022	0.0092	0.0375	0.246	0.8054	[-0.0642, 0.0827]
Year***	0.0324	0.0043	7.452	0.0000	[0.0239, 0.0409]

N = 7,583 R^2 = 0.0724
Formula: log_total_wages ~ post2022 + C(EmployerName) + Year | Research professors only.
HC1 heteroskedasticity-robust standard errors. Stars: *** p<0.001 ** p<0.01 * p<0.05

Figure 15: Within-Rank Regression: Research Professors. $N = 7{,}583$. The post2022 coefficient is positive and not statistically significant ($p = 0.805$), indicating no significant deviation from trend within this smaller rank category.